

The beginnings of a new era: time to reflect on 17 years of the ISJ

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Abstract. *In this paper we reflect on the first 17 years of the Information Systems Journal (ISJ). The reflections are considered under three headings: origin of papers (authors, geographical period, gender and departments), research paradigms (positive/interpretive, qualitative/quantitative, non-empirical/empirical, espoused theories and research method) and finally, topics. We find that throughout the period, the published papers evidenced a greater internationalization of the journal. On the other hand, some regions and countries are poorly represented. Another imbalance concerns the lack of practitioner papers along with an author-gender imbalance. Qualitative research exceeds quantitative research by a factor of more than 2:1. Interestingly, papers classified as descriptive/conceptual/theoretical have been largely superseded in the period by those that have some empirical evidence to illustrate the points made. Sometimes it is argued that the discipline of information systems lacks theory and thinking pieces but the ISJ suggests that this is far from the truth. Further, our analysis has revealed around 250 topics discussed using many research methods to explain the phenomena. Depending on the point of view, this may show that the discipline either lacks focus or is exciting and pluralistic. We lean to the latter view, but others have argued for focusing on fewer topics and research methods.*

Keywords: information systems, journals, theories, topics, research approaches

BACKGROUND

This first issue of Volume 18 of the *Information Systems Journal (ISJ)* marks another milestone – four in fact – of the journal's progress. First, we have moved from four to six issues annually. This reflects that the journal is now attracting a larger number of high quality submissions. It has been doing so for some time but we have waited to make this move until we were convinced that the expanded content was sustainable in terms of quality papers. Second, from this issue, the journal receives the official approval of the *Association of Information Systems (AIS)* – hence the AIS logo on the front cover. The AIS represents the international information systems academic

Table 1. *ISJ* published papers in three groups to analyze trends*

Group	Volume numbers	Numbers of papers	Percentage of papers (%)
1	1–6	95	34.55
2	7–12	96	34.91
3	13–17	84	30.55

*In our descriptive and discussion-oriented review we have provided percentages only in our tables. Full data, including raw numbers and the results of Chi-Square, df and Asymp. Sig. tests, can be provided on request by emailing David Avison (Avison@essec.fr).

community (see www.AISnet.org and www.AISworld.org) and is the leading international organization for information systems (IS) teaching and research. We are gratified by this important recognition, at this time unique to a journal not published by the AIS itself. Third, for the first time, *all* the past papers in *ISJ* (rather than most) will soon be available for downloading (see Synergy at <http://www.blackwell-synergy.com/loi/isj> or your library's journal database). Along with the publisher's Web site, we as Editors have our own Web site at <http://www.disc.brunel.ac.uk/isj>. Finally, the *ISJ* is included in the AIS *basket* of six journals which the AIS suggests are the top journals in our field (see <http://home.aisnet.org/displaycommon.cfm?an=1&subarticlenbr=1>) following the recommendations of the IS Senior Scholars.

We feel that it is therefore both opportune and useful to look back and reflect on the previous volumes and history of *ISJ* as we make this important 'step forward'. This is not the first time we have done such reflecting and in 2001, when the journal was 10 years old, we published *Reflections on information systems practice, education and research: 10 years of the Information Systems Journal* (Avison et al., 2001). Now, with 17 years' experience we feel that it is appropriate to analyze and reflect on the development of the journal and the changes and trends that have occurred during this time.

Although this paper addresses issues concerned with the content of the journal and our policies as Editors of the *ISJ*, we look briefly at readership. More recently, download figures have taken predominance over hardcopy subscriptions. Paper downloads amounted to 30 000 in 2003, 46 000 in 2004, 57 000 in 2005 and over 70 000 in 2006. Year-to-date figures suggest that this steep upward trend will continue in 2007. Although hardcopy subscriptions are small in comparison, the publishers intend to continue the paper edition of the journal. One reason is that in general, Deans rate journals that have a hard copy edition higher than those that are electronic only (for whatever reason). Another key figure is impact factor. The latest figure (for 2006) showed a marked increase to 1.543.

The approach we have adopted for this study is to analyze the whole 17 years of the *ISJ*¹ and then we have broken down the analysis into three roughly equal groups in order to help identify trends. Table 1 shows the composition of these three groups in terms of number of papers and percentages for each group.

Unlike computer science (CS), which focuses on the technology, we have always seen IS as looking outward from the technology and towards the organization and society. The previous

¹In this analysis we have omitted the special issue in volume 17 devoted to celebrating the work of Enid Mumford.

'reflections' paper demonstrated that IS influenced and was influenced by a number of related disciplines and it embraced pluralistic research methods. In this paper we look first at the origins of papers, then at research paradigms and theories used by researchers, and finally at topics covered.

ORIGIN OF PAPERS

We begin our analysis by looking at the origins of the papers submitted and published over the period, in relation to number of authors, geographical regions, gender and departments. There have been 437 different authors and 548 contributors (the difference being explained by authors who have contributed more than one paper). Of the papers published, 32.4% were single authored, 44.4% (which form the largest category) were by two authors, 17.8% by three authors and 5.1% by four authors or more. For reasons difficult to explain, single authored articles appear to be decreasing over time as 43.2% single author articles appeared in issues 1–6 but this reduced to 28.1% in issues 7–12 and further reduced to 25% in issues 13–17. It is surprising because many promotion and probation (tenure) committees still favour single authored papers. Perhaps academics in the field are becoming more collaborative.

In terms of the number of contributors from different geographical regions, most (33.4%) were located in North America, closely followed by the United Kingdom, with 32.5%. The third largest category (19.7%) was formed by European (non-UK) contributors followed by the Asia-Pacific region with 13.3% contributors. The remaining 1.1% ($n = 6$) contributors were from outside these broad regions. It is a concern to us that we seem to be missing out from a large part of the world including South America, Africa and, to a lesser extent, China, and whilst there has been historically relatively little IS research in these areas known in the 'West'. However, this trend is certainly changing, and indeed one of our Associate Editors (AEs), Robert Davison, is currently editing a special issue on IS in China which will be published shortly. However, we still need to think how we can address IS in these under-represented geographical areas.

Table 2 illustrates that the contributors from the UK have decreased over time as 41.9% authors from the UK contributed articles within issues 1–6, but this reduced to 19.2% within issues 13–17. In contrast, the trend suggests that the number of contributors from the other main geographical regions including North America, Asia-Pacific and mainland Europe have

Table 2. Geographic location of papers

Geographic region	Volume 1–6 (%)	Volume 7–12 (%)	Volume 13–17 (%)
Asia-Pacific	13.2	13.7	13.0
Europe	14.4	19.1	25.4
North America	29.3	29.9	41.2
UK	41.9	36.3	19.2
Other	1.2	1.0	1.1

increased over time. This is perhaps not surprising as the *ISJ* was created and established in the UK with UK Editors, and in its early days attracted many UK submissions, although, as can be seen, it was never purely a UK journal. Since then it has constantly improved its international audience and reputation. Indeed *ISJ* has been identified as one of the most international IS journals (Galliers & Meadows, 2003), significantly more so than other journals, such as *Management Information Systems Quarterly (MISQ)* and *Information Systems Research (ISR)*.

Based on an analysis of authors' institutions, contributors came from 29 different countries, with the UK (32.7%) and the US (31.2%) dominating. The third largest category (7.8%) was formed by Australian authors, followed by Denmark (3.3%). Whereas, authors from the UK have declined over time, the number of authors from the US, Denmark, Taiwan, Finland, Greece, Hong Kong, France, Austria and Israel have all increased.

In terms of the number of contributors from academia or industry, the vast majority (92.9%) were from the academic world with only a small proportion from industry (7.1%). This has been fairly consistent throughout and is a disappointment to the Editors. When the journal was established, it was hoped that it would be more of a mixed journal. However, we can only repeat what we said in our earlier 'reflections paper', that it is 'unfortunate that most papers submitted by practitioners as sole authors fall foul of our stringent refereeing process. This causes us some concern given our belief in the importance of practice' (Avison *et al.*, 2001). We are not the only ones to recognize this and as has been observed, there does seem to be 'Disconnects between the world of Business and Academia' (Hirschheim & Klein, 2003). Even though many *ISJ* papers are based on case studies and action research, reflecting research in organizations, attracting practitioner papers does seem to be problematic, and very few publications are able to bridge that gap successfully.

A gender imbalance also concerns us. The fact that the percentage of female authors has increased from 14.4% (volumes 1–6) to 19.8% (volumes 13–17) is a move in the right direction but it still indicates a structural problem; probably a reflection of a wider issue which is the gender imbalance in universities, including in our domain.

Analysing the information provided by authors as to the departments in which they are located, we find (Table 3), perhaps unsurprisingly, that the largest number of contributors are located in departments of IS (or similar title; 43.4%), which is followed by Business/Management departments (34.9%), and then by CS departments (11.1%). This suggests that IS academics may be more research active in independent departments than in IS groups within a CS department. IS academics in business or management departments seem to be more successful. Other departments contributed very few articles.

As for trends, contributors from CS and other reference disciplines except Business/Management have decreased over time as 22.8% authors from CS contributed articles within issues 1–6 but this reduced to 9.3% within issues 7–12 and further reduced to 2.8% within issues 13–17. In contrast, the number of contributors/authors from the IS and Business/Management categories have increased over time. Thus, it seems that the *ISJ* is becoming more IS with a management focus than IS with a CS one, and maybe this reflects the way that the subject area as a whole has evolved with a move away from the technical issues towards the application and effectiveness issues within organizations and society.

Table 3. Home departments of authors

Author home departments	Volume 1–6 (%)	Volume 7–12 (%)	Volume 13–17 (%)	Total (%)
Business and management	38.9	33.3	32.8	34.9
Computer science	22.8	8.8	2.8	11.1
Information systems	28.1	46.1	54.8	43.4
Sociology	0.0	1.0	0.0	0.4
Statistics	0.0	0.5	0.0	0.2
Psychology	0.0	0.5	0.0	0.2
Engineering	1.2	1.5	1.1	1.3
Medicine	0.6	0.0	0.0	0.2
Others	1.2	0.0	2.8	1.3
Practitioners	7.2	8.3	5.6	6.9

Avison & Ein-Dor (2007) have argued that a trend towards more IS academics being in their own departments (rather than within business or CS departments) is desirable. This trend also points to the increasing separation of IS from CS. Notwithstanding this trend, in the UK at least, university funding bodies have supported the 'harder', more technical aspects of CS more than the 'softer' organizational and people aspects.

Analysing the stated affiliations of authors, we find that 240 different universities/organizations have been represented, with Georgia State University providing the largest number of contributors (27). The London School of Economics and Political Science is second with 17 contributors. The University of Queensland and the University of Warwick both had 14 contributors, whilst 11 contributors were from the University of Salford. The University of Manchester Institute of Science and Technology and the University of Manchester both had 10 contributors (these two institutions have now merged so if the future reflects past trends we might expect the merged the University of Manchester to be more prominent). Nine authors/contributors were affiliated with University College Cork. Eight authors/contributors were affiliated with the following three universities: Brunel University (the 'home' of the *ISJ*), Copenhagen Business School and University of Oxford. From the remaining numbers, seven universities had affiliation with seven authors from each and five universities had affiliation with five authors from each. Following this, 16 universities had four contributors, 15 universities had three contributors and 43 universities had two contributors from each. Finally, the largest number (143) of universities had affiliation with just one author from each. One reflection on this data is that the *ISJ* has not been dominated by a particular university or author style, in other words, its papers are not written to a certain formula, an accusation made in regards to some journals.

The success of the *ISJ* has as much to do with the quality of our Associate Editors and referees as to the authors. We are often particularly impressed to see the huge improvements in an article from original submission to final publication. All *ISJ* papers are screened by the Managing Editor and around half the originally submitted papers are returned to authors, because they need further refinement before resubmission, because they do not address the scope of *ISJ* or they are simply not up to the required standard. Associate Editors presently

based in the Australia, Denmark, France, Hong Kong, Ireland, Singapore, Spain, the UK and the US generally assign referees from their own region to review papers. All papers have two or more reviews unless the Associate Editor determines that the paper is unsuitable. The geographical imbalance of paper submissions discussed earlier implies that more papers are reviewed in Asia/Oceania than are submitted from these regions while fewer are reviewed in certain areas, especially in the US. The distribution of referees reflects more the distribution of Associate Editors than the origin of authors. Twenty-five percent of reviewers are based in the UK, 30% in mainland Europe, 33% in North America, and 12% in the Asia-Pacific region.

No papers are accepted without some changes, very few with only minor changes on the first review cycle. Most papers cycle through the system at least twice plus a final tweak. Special issues follow a slightly different route with guest editors overseeing the process, which is equally rigorous. Our review processes mean that all *ISJ* papers tend to go through a number of review cycles. The average number of review cycles is 3.2, with no recent paper undergoing fewer than two cycles and some going through the mill five times. The median number of cycles is three. Most papers are reviewed and re-reviewed by two referees in addition to the Associate Editor and the Managing Editor. However, around 8% of papers have three reviewers, usually when the first two produce very differing responses.

As Editors, we have often found contrasting reviews for a paper 'interesting' and such a paper is often sent for another review. We do not immediately reject a paper if at least one of the reviews is strongly positive, when full reasoning for such a view is given. We would prefer to pursue this than one that gains a consistently 'mildly positive' review, the danger being that a boring paper which adds little if anything to knowledge in the IS community is accepted almost by stealth.

Despite (or indeed because of) this thorough review process, we think it important to give authors a good experience when submitting to the *ISJ*. A study by Bhattacharjee *et al.* (2004) on author experiences of the review process ranked *ISJ* very highly in all categories. It was ranked third, out of 39 journals, for the 'Quality of its Reviews' (first out of the IS journals). It was ranked first overall for the 'Responsiveness of its Editorial Office' and *ISJ* was found to be seventh out of 39 for the time papers spent in the review process, with an average for first reviews of 3.4 months and 2.8 for second rounds of reviews. These figures are around half of the average of all the journals and significantly better than some other competitor journals where papers spent over 18 months in the review process, on average. However, this record is thanks to our reviewers, AEs and our editorial office at Brunel University. Sometimes we do not meet our exacting standards as referees and other colleagues are not paid for this work and have great demands on their time.

RESEARCH PARADIGMS

Having looked at some demographic information concerning authors we now turn to look at the papers themselves. We begin by examining the general research paradigm used in our published papers. Initially, we categorize the papers into two basic groups: positivist and

Table 4. Types of research

Positivist vs. interpretive	Volume 1–6 (%)	Volume 7–12 (%)	Volume 13–17 (%)	Total (%)
Positivist	23.2	32.3	32.1	29.1
Interpretive	76.8	67.7	67.9	70.9

interpretive. In undertaking the analysis we had to be somewhat interpretive ourselves, as authors do not always specifically categorize their work in this way. In making such decisions, we followed Chen & Hirschheim's (2004) criteria for categorizing positivist articles as such if they had indications of hypotheses, propositions, model formation, quantifiable measures of variables and the inferences drawn from samples to populations (Orlikowski & Baroudi, 1991), and demonstrated the belief that 'objective' data could be collected to predict the relationship among factors and to test hypotheses or theories (Walsham, 1995).

The criteria for determining interpretive articles were threefold. First and foremost, articles should not involve any positivist indicators as described above, that is, no deterministic perspectives imposed by the researchers. Second, participants' perspectives are taken as the primary sources of understanding and investigating the phenomena. Third, the phenomena are examined with respect to cultural or contextual circumstances (Walsham, 1995).

As can be seen in Table 4, the *ISJ* has published predominantly interpretive research, with around 71% of papers categorized in this way, and 29% categorized as positivist. This is no surprise as one of the original objectives of the Editors (David Avison and Guy Fitzgerald) when they proposed the journal to the publishers in the late 1980s was to provide a natural home for interpretive research papers. We felt that such research was not well catered for in terms of outlets at that time. However, within the groups there is a small indication of an increase in the number of positivist papers over the initial period, which is slightly surprising, although it seems to have settled to a steady 70/30 split. It was never the intention to publish interpretive papers only as the Editors believed in a pluralism of research methods and paradigms as being appropriate for the relatively young subject area. Over the years since the creation of the journal there have been a number of debates, and indeed proposals, that it become a purely interpretive journal. This has been rejected because of our pluralist views, and the *ISJ* continues to welcome the best of positivist research.

Given the predominance of papers in the interpretive classification (71%) we look at these in more detail and further refine the analysis. Our first sub-category of interpretive includes papers that are essentially empirical and comprise 42% of the interpretive total. The criterion that we used for categorizing studies as empirical or non-empirical is based on whether the articles obtain data or observations, which could be gathered through quantitative, qualitative or a mixed approach, including archival data. Empirical studies rely on observations and data, whereas non-empirical studies emphasize ideas and concepts (Alavi & Carlson, 1992).

The next sub-category is critical research which adopts an evaluative dimension (Orlikowski & Baroudi, 1991). The critical researcher attempts to critically evaluate and transform the social reality under investigation. Where interpretive research is content to predict or explain the

Table 5. Research paradigm/epistemology of interpretive papers

Interpretive papers	Volume 1–6 (%)	Volume 7–12 (%)	Volume 13–17 (%)	Total (%)
Interpretive-empirical	23.3	55.4	49.1	41.5
Critical	0.0	1.5	5.3	2.1
Descriptive/conceptual/theoretical	76.7	43.1	45.6	56.4

status quo, 'the critical perspective is concerned with critiquing existing social systems and revealing any contradictions and conflicts that may inhere within their structures. Through fostering this type of self-consciousness and understanding of existing social conditions, critical researchers believe they can help to overcome oppressive social relations' (Orlikowski & Baroudi, 1991, reproduced in Richardson & Robinson, 2007). As can be seen (Table 5), there were relatively few critical papers (only 2.1%), but this will increase as there are a numbers of critical papers in the *ISJ* pipeline (thanks especially to the special issue edited by Heinz Klein, Dubravka Cezec-Kecmanovic and Carole Brooke).

The other sub-category of interpretive research is Descriptive/Conceptual/Theoretical and these were essentially the non-empirical papers, composed of either descriptive (or argumentative) approaches, conceptual or theoretical deliberations. As can be seen, these make up 56% of the interpretive papers overall. However, the trend indicates a significant decline from the early period (volume 1–6) when 77% of the papers belonged to this category to around 45%, which we interpret partly as a reflection of the maturity of the discipline. In the early years we were, to some extent, searching for an understanding of the foundations of the discipline and so were particularly concerned with theory building. Indeed, this was one of the main thrusts of our paper published in the first issue (Avison & Fitzgerald, 1991). However, we also think it reflects changing expectations and tougher requirements of our referees. Unless the ideas and arguments expressed in a paper are particularly strong and attractive, our referees are now saying: 'yes, but give us the evidence!'

This evidence is largely being provided by empirical research with an increase in such papers from around 23% of interpretive papers in the first period to 43% in the second and 45% in the latest period; with a reciprocal decrease in papers based on non-empirical research over the same period. Again, following Chen & Hirschheim (2004), we argue that while non-empirical studies help to develop concepts and build theory, empirical studies provide concrete evidence for testing theories. As a discipline becomes more mature, one might expect theory-testing to outweigh theory-building efforts. Therefore, a comparison between empirical and non-empirical studies serves as a meaningful indication of the progress made in this context.

It is not, to us, surprising that the proportion of interpretive research has grown much faster than positivist research, and interpretive research is seen as a particular strength of the *ISJ*. Nevertheless, the Editors' view is that the most important aspect is that the research method chosen must be appropriate for the research question and also employed correctly. Whatever the research method used, over this time the evidence of research quality has become more apparent through papers having more focus, thought, rigour and explanation of the adopted research method.

Table 6. Theories and theoretical constructs

Actor-network theory	McDonaldization theory
Agency theory	Metaphors
Appreciative inquiry	Narrative thinking
Bargaining theory	Nolan's stage theory
Behavioural decision theory	Norm-based agency
Chaos theory	Organizational learning
Capability maturity model	Organization theory
Contingency theory	Personal construct theory
Critical thinking	Popperian philosophy of science
Critical social theory	Pragmatics
Economic theory	Quinean philosophy of science
Ethical theory	Resource-based theory
Feminine theory	Sense making
Game theory	Social action
Giddens' structuration theory	Social presence theory
Habermas's communicative theory	Socio-technical theory
Hackman's model of group effectiveness	Speech act theory
Human cognition	Stages of growth model
Information theory	Stakeholder theory
Innovation diffusion	Systems theory
Institutional theory	Theories of discourse comprehension
Kohlberg's theory of moral development	Theories of meaning
Lewin's change theory	Theories on motivation and training
Logico-linguistic model	Theory of planned behaviour
Masculine theory	Upper echelon theory

From our categorization of Descriptive/Conceptual/Theoretical papers, we identify 50 theories and theoretical constructs used by authors over this period (Table 6) from the list of 231 topics (see Table 10). Although it might be argued that many of these are not fully-fledged theories, it seems to counter the much-professed argument that IS lacks theory, thinking and ideas. It also could be said to indicate that the *ISJ* in particular has been at the forefront of 'thinking' in the discipline. Yet the list might also be used as a weapon by those that argue that the discipline lacks focus. Many, but not all, of these theories have been developed in other disciplines, but we see no valid argument that IS should not benefit from other disciplines, as indeed it itself influences and benefits other disciplines, as Baskerville & Myers (2002) demonstrate.

The next step of the analysis relates to the research method adopted in the papers (if the paper is of the type where a research method is appropriate and one is adopted, as clearly some papers, particularly those in the Descriptive/Conceptual/Theoretical category, may not have a research method). The analysis began by asking whether the research was quantitative or qualitative. Again, we utilize the definitions of Chen and Hirschheim (2004) here and classify papers as quantitative if they use numerical analysis to illustrate the relationship among factors in the phenomenon studied, and we define it as qualitative if the emphasis is on the description

Table 7. Quantitative vs. qualitative papers

Quantitative/qualitative	Volume 1–6 (%)	Volume 7–12 (%)	Volume 13–17 (%)	Total (%)
Quantitative	17.9	16.7	20.2	18.2
Qualitative	22.1	43.8	42.9	36.0
Mixed	1.1	10.4	6.0	5.8
Descriptive/theoretical/meta analysis	58.9	29.2	31.0	40.0

and understanding of the situation behind the factors. It is possible that a paper could use both quantitative and qualitative methods, maybe in different stages of the research, in which case we categorize them as mixed. These three categories (quantitative, qualitative and mixed) essentially relate to papers that are based on empirical research (see also, e.g. Avison & Myers, 2005; Straub *et al.*, 2005). Of course there are a range of papers that are non-empirical and somewhat diverse that we categorize as Descriptive/Theoretical/Meta Analysis and non-empirical studies typically involving, describing or arguing for theories and frameworks. Table 7 shows the outcome of this analysis and we see that around 40% are classified as Descriptive/Theoretical/Meta Analysis, with 36% qualitative and the rest either quantitative or mixed.

It will be noted that there are parallels between this analysis and the positivist/interpretive discussion above and that the figures are not dissimilar to the quantitative/qualitative percentages.

We are somewhat surprised at the relatively low proportion of papers published that use both quantitative and qualitative approaches, i.e. mixed, as we thought that a typical IS paper often involves an exploratory survey (i.e. quantitative) to establish some broad findings followed by a more in-depth series of interviews or case studies to create some understanding and explanations of some aspect of the survey (i.e. qualitative). Clearly, this kind of research is less common than we thought with only a total of around 6% in this category. Perhaps, in these days of publication pressures, such research is published as two separate papers rather than one!

We now turn to research designs evident in the papers. We have partly adopted the classifications of Orlikowski & Baroudi (1991) in this context, these are: survey, case study, laboratory experiment, field experiment, action research, ethnography, grounded theory and 'others'. Survey studies typically gather data by means of questionnaires. Case studies involve research in an organization, either single or multiple site, over a certain period of time and such inquiries usually rely on 'multiple sources of evidence' (Yin, 1994, p. 13). Laboratory experiments include studies that take place within a designed, controlled environment and usually involve special treatments of different groups to contrast the precise relationships among variables (Galliers, 1991). In field experiments, as with laboratory experiments, special treatments are used to study two or more controlled groups. Field experiments differ in that they are conducted in a real-world setting. Action research is often embodied in a case study, but distinguishes itself in that the researchers are an integral part of the phenomenon under study. The researchers' input often influences the outcomes of the phenomenon and his or her role could change from researcher to subject (Galliers, 1991). The 'others' category includes articles that are practitioner-oriented (systems or tools development), non-empirical pieces or

Table 8. Research method (adapted from Palvia & Pinjani, 2007)

Research method	Volume 1–6 (%)	Volume 7–12 (%)	Volume 13–17 (%)	Total (%)
Survey	8.4	10.4	16.7	11.6
Case study	10.5	29.2	29.8	22.9
Laboratory experiment	4.2	4.2	4.8	4.4
Field experiment	2.1	0.0	0.0	0.7
Action research	4.2	7.3	10.7	7.3
Ethnography	0.0	2.1	1.2	1.1
Grounded Theory	0.0	0.0	1.2	0.4
Interview	4.2	9.4	1.2	5.1
Speculation/commentary	8.4	1.0	3.6	4.4
Library research/literature analysis/frameworks/ conceptual modelling	50.5	26.0	25.0	34.2
Mathematical model	1.1	0.0	1.2	0.7
Secondary data analysis	1.1	3.1	0.0	1.5
Content analysis	1.1	1.0	0.0	0.7
Multi-method	2.1	5.2	3.6	3.6
Others	2.1	1.0	1.2	1.5

descriptive/argumentative as noted in Galliers' (1991) classification. Research with secondary data such as public records or existing datasets is also located in this category.

As seen in Table 8 (adapted from Palvia & Pinjani, 2007), we classify the papers into a total of 15 categories based on their research method. The findings indicate that the largest number of articles was in the Library Research/Literature Analysis/Frameworks/Conceptual Modelling (34.2%) category, with Case Study research (22.9%) second, followed by Survey research (11.6%), Action Research (7.3%) and Laboratory Experiments (4.4%). Apart from these, all other categories were employed by very few articles, which was somewhat surprising to us, as IS is usually thought to be a highly pluralistic discipline. Perhaps this suggests that it is less diverse than is sometimes assumed, at least in the research methods used, though it could also reflect the bias of our editorial and reviewing system. For example, the *ISJ* has always been reticent to publish research that uses students as inappropriate surrogates for other populations in laboratory experiments.

TOPICS

We have adapted Barki *et al.* (1993) for the analysis of IS research topics (Table 9). The findings suggest that the largest number of articles investigated research issues related with the IS Development category (30.5%), closely followed by the IS Management category (28.7%). The third most researched category was IS Education, as 36 articles (13.1%) fell within this category, followed by an almost equal number of two categories, namely IS Types

Table 9. Categories (basedon Barki *et al.*, 1993)

Major categories in IS research	Volume 1–6 (%)	Volume 7–12 (%)	Volume 13–17 (%)	Total (%)
Reference disciplines	2.1	2.1	3.6	2.5
External environment: economic, legal, political and social	1.1	1.0	3.6	1.8
Information Technology, computer systems and software	1.1	0.0	0.0	0.4
Organizational environment, including characteristics, function	2.1	13.5	8.3	8.0
IS management, including management of hardware, software, personnel, project, etc.	26.3	34.4	25.0	28.7
IS development and operations, life cycles, IS development	38.9	29.2	22.6	30.5
IS usage, by organizations and users, and support, access	7.4	4.2	7.1	6.2
IS types, application areas, components and characteristics	8.4	7.3	10.7	8.7
IS education and research	12.6	8.3	19.0	13.1

IS, information systems.

and Applications (8.7%) and Organizational Environment (8%). The IS Usage category was also represented by some articles (6.2%). The remaining three categories were represented by very few articles.²

Trends from the analysis suggest that the number of articles from the majority research category (IS Development) has decreased over time, from around 39 to 23%, which is interesting. It seems to indicate that one of the original core IS research areas is in decline, at least from a research perspective. Perhaps all those IS development issues, such as on-time and within budget, that have plagued the industry since its inception, have been at least partly overcome? Perhaps it is a matter of fashion? Or maybe such research has migrated more into CS journals? Or perhaps the outsourcing of significant amounts of systems development means that the issues are not so much in the research forefront as they once were. Whatever the reason, it seems like a significant change. Topics which have increased to compensate include IS Education, IS Types, Organizational Environment, and IS Usage. For example, 2.1% of articles from the Organizational Environment category appeared within issues 1–6, but this increased to 13.5% within issues 7–12 and the lower percentage of 8.3% within issues 13–17.

Table 10 takes the analysis of research areas further from the broad categories of Table 9 and identifies a range of specific topics, theories and theoretical constructs utilized by authors. Many of these fall into the category of descriptive/conceptual/theoretical papers. As can be seen, a large number were identified, nearly 275! The problems of such a wide range of topics, theories and constructs has been recognized before and has led some to suggest that the discipline faces an ‘anxiety discourse’ or ‘identity crisis’ (see, e.g. papers in King & Lyytinen, 2006) and it is understandable that Benbasat and Zmud (2003) and others have suggested focusing more on central core themes in IS. However, the *ISJ* has always reacted against this view, as we see it as restricting the potential of the discipline (this is not an excuse for a lack of rigour!). But it is partly because of this wide range of topics, theories and constructs that make IS such a rich and

²In our review we have not provided a paper-by-paper categorisation. Such data can be provided (Avison@essec.fr).

Table 10. Topics, theories and theoretical constructs

Accountability	Designer accountability	Information needs
Actor-network theory	Diffusion of innovation	Information seeking
<i>Ad hoc</i> workgroups	Diffusion models	Information sharing
Adaptation	Digital divide,	Information society
Agency theory	Discourse analysis	Information systems quality
Allegiance	Disintermediation	Information theory
Ambiguity	Disruptive innovation	Information use
Appreciative inquiry	Double-loop learning	Infusion
Argumentation	Economic theory	Innovation diffusion
Authoring	Economic transformation	Institutional theory
Autopoiesis	Effectiveness	IT/IS investment
Autonomy	Electronic community	Job satisfaction
Bargaining theory	Empowerment	Kohlberg's theory of moral development
Beer game role playing	Evolving information systems	Laddering
Behavioural decision theory	Ethical attitudes	Language game
Bureaucracy	Ethical theory	Legitimization
BPR	Face-saving	Lewin's change theory
Business strategy	Fashion setting	Logico-linguistic model
Capabilities	Falsificationism	Masculine theory
Channel disposition	Feminine theory	McDonaldization theory
Changing economies	Flexible specialization	Media richness
Change management	Game theory	Mediation
Chaos theory	Genealogy	Metaphor
Capability maturity model	Giddens' structuration theory/ paradigm	Mum effect
Cognitive structure	Gift giving	Narrative thinking
Collaboration	Gift economy	Nolan's stage theory
Computer ethics	Gift culture	Norm-based agency
Competitive advantage (also as sustainable competitive advantage)	Globalization	Novice users
Competence	Grounded theory	Organizational change
Consonance	Critical social theory	Organizational context
Conflict	Habermas's communicative theory	Organizational co-ordination
Conflict management	Hackman's model of group effectiveness	Organizational culture
Consequentialism	Hierarchies	Organizational form
Consumer behaviour	Holism	Organizational impacts
Contextualized access	Holistic management	Organizational learning
Continental philosophy	Human cognition	Organizational leadership
Contingency theory	Human information processing	Organizational performance
Contracting	Hybrid managers	Organizational strategies
Constraint	Implementation commitment	Organization structure
Cooperative group work	Implementation success	Organization theory
Cost-benefit analysis	Incomplete contracting	Organizational vision
Core competencies	Individual decision making (also as decision-making)	Outsourcing
Critical thinking	Information assets	Ownership
Critical social theory	Information competencies	Participation
Culture	Information environment	Participative design
Data quality		Partial adoption
Deontology		Partnership

Table 10. Cont.

Personal construct theory	Self-efficacy	Success
Perceived resources	Self perception	Systems engineering
Phenomenology	Self-regulation theories	Systems theory
Playlets	Semantics	TAM
Politics	Semiotics	Team
Popperian philosophy of science	Sense making	Team performance
Power	Situated learning	Theories of discourse comprehension
Pragmatics	Skills/skill development	Theories of meaning
Privacy	Sleep deprivation	Theories on motivation and training
Process management	Social action	Theory of planned behaviour
Process tracing	Social behavioural science	Total systems intervention
Productivity,	Social defence	Transaction cost economies
Professionalization	Social determinants	Transparency
Project management	Social influence	Trust
Public administration	Social integration	Upper echelon theory
Quality assurance	Social sociology of technology	Uncertainty
Quality improvement	Social process	Universal grammar
Quality management	Social presence theory	User behaviour
Quality of information	Social psychology	User satisfaction
Quinean philosophy of science	Social struggle	Value creation
Rationalization	Social systems	Value focused thinking
Relativistic nature	Socio-economic context	Vickers' notion of appreciation
Resource-based theory	Socio-technical theory	Virtual positions
Reflective practice	Social work	Virtual teams
Reputation	Software process maturity	Virtual teamwork
Risk	Sourcing	Virtual organization
Risk mitigation	Speech act theory	Visible systems model
Role ambiguity	Stages of growth model	Whistle blowing
Role conflicts	Stakeholder theory	Work and mobility
Security vulnerabilities	Strategy	Work-organization
	Strange attractors	
	Strategic change	

BPR, business process reengineering; TAM, technology acceptance model.

exciting discipline to research. We are rather sympathetic to Weber's (2003) argument following Darwin that the good ideas will last while the weak ones will disappear.

FURTHER REFLECTIONS

In this review paper reflecting on 17 years of the *ISJ*, we see some trends that apply to this journal and may apply to the discipline as a whole. Looking first at the origins of papers published, surprisingly, in view of the kudos likely gained by the author, the proportion of single-authored papers has declined in the period by almost half. We would hope this is due to the perceived advantages of greater collaboration. It is less surprising that authors from the UK have

decreased by over half during the period. This reflects the greater internationalization of the journal. On the other hand, although North America, Europe and Asia-Pacific are well represented, it is disappointing that so few articles emanate from China and India, which are hotbeds of IS practice and becoming hotbeds of IS research, along with South America and Africa. We hope to address this imbalance and have a special issue on IS research in China in the pipeline. However, our attempts to attract excellent practitioner papers have largely failed with only 7% of articles coming from industry. The gap between practice and research has not been filled by the *ISJ*, or indeed any other highly rated research journal and conference. Another imbalance concerns the gender of authors, as even now only around 20% are women. Interestingly, most of our authors came from either university departments of IS or similar or business/management departments rather than CS ones which are also likely homes for our IS academic colleagues. This could reflect the movement of IS research away from technical to organizational and people issues, though it might also reflect our editorial bias. Finally, as we look at the origins of papers, we see that although some university departments are quite well represented, the *ISJ* has not been dominated by a particular university or author style, suggesting we aim at diversity rather than 'formulaic' papers. We are also aware that many university departments include the *ISJ* as a top ranking research journal (the AIS Web site also lists the *ISJ* amongst its top six ranking of IS research journals). This high ranking reflects the quality of the Board of Editors, Associate Editors, reviewers and authors of the papers published.

In terms of the research paradigms used by authors in the published research, the *ISJ* is predominantly an outlet for interpretive rather than positivist research, and qualitative rather than quantitative, by a factor of over 2:1 in both cases. This reflects the espoused aims of the journal 17 years ago, which went against the general trend of the time for research journals. Indeed we set up the *ISJ* originally precisely to provide an outlet for such research. Thanks to groups such as the Information Federation of Information Processing Working Group 8.2, as well as the *ISJ*, interpretive and qualitative research is now more mainstream but still not equal in journals such as *MISQ* and *ISR*. Nonetheless, around a third of our papers do not fall into these categories and this reflects our pluralistic attitude, wishing to publish the best of IS research, whatever its research paradigm. Interestingly, papers that have been classified as descriptive/conceptual/theoretical have been largely superseded in the period by those that have some empirical evidence to illustrate the points made, perhaps reflecting a maturing of the discipline. Of the interpretive research methods used, we see critical research being more recently apparent (indeed there is a forthcoming special issue on this 'missing' paradigm). We are surprised to see design research rarely being explicitly mentioned, though on further analysis of the papers, we see this research approach as being implicit in some, particularly those discussing IS development. Sometimes it is argued that the discipline of IS lacks theory and thinking pieces. Judged by the *ISJ*, this is far from the truth! Further, our analysis has revealed around 250 topics discussed using many research methods to explain the phenomena. Depending on the point of view, this may show that the discipline either lacks focus or is exciting and pluralistic. We lean towards the latter view, but others have argued for focusing on fewer topics and research methods. We wonder who will 'win' the argument, and look forward to analyzing the *ISJ* as it reaches its 25th birthday in 2016!

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